The three-body problem – from Newton's headache to chaos

Two bodies attracting each other move in ellipses – Newton solved that completely. Add a third, and the problem has no general solution in formulas. The three-body problem has shaped mathematics for three hundred years: it gave us the Lagrange points, where the James Webb telescope floats today, Poincaré's discovery of chaos, and a figure-eight orbit first found in 1993. Here is the story – with a simulator in which you can release three bodies yourself, and the landscape of the Lagrange points in 3D.

Two bodies: solved

For two bodies of masses m_1 and m_2 the motion can be split into the uniform motion of the centre of mass and the relative motion $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$, which satisfies

$$\ddot{\mathbf{r}} = -\frac{G(m_1 + m_2)}{r^3}\,\mathbf{r}.$$

This is a problem of one body in a fixed central field, and Newton showed in the Principia that the orbit is a conic section – an ellipse, parabola or hyperbola. The problem has enough conserved quantities (energy, angular momentum and the Laplace–Runge–Lenz vector) to determine the orbit completely. With three bodies there are 18 unknowns (three positions and three velocity components in space for each body) but only 10 classical conserved quantities: the energy, three components of momentum, three of angular momentum, and the three initial coordinates of the centre of mass. Heinrich Bruns showed in 1887 that there are no further algebraic conservation laws. The equations

$$m_i \ddot{\mathbf{r}}_i = \sum_{j \neq i} \frac{G\, m_i m_j\, (\mathbf{r}_j - \mathbf{r}_i)}{|\mathbf{r}_j - \mathbf{r}_i|^3}, \qquad i = 1, 2, 3,$$

therefore cannot be reduced to something that can be solved by formulas.

Newton and the Moon

Newton met the problem at once, because the Moon does not move in an ellipse. The Sun pulls on both the Earth and the Moon, and the Moon's orbit turns, wobbles and changes shape (see The Moon's orbit in 3D). In Book I of the Principia, Proposition 66 with its 22 corollaries, Newton described qualitatively how a third body disturbs an ellipse, and in Book III he tried to calculate the Moon's

inequalities. He obtained the rotation of the line of apsides only half as large as it is. According to an account from Newton's circle, he told Halley that the theory of the Moon made his head ache and kept him awake so often that he would think of it no more. The problem was solved only in 1749, when Alexis Clairaut realised that second-order terms in the perturbation calculation must be taken into account.

The simulator: release three bodies

Here Newton's equations are solved numerically with fourth-order Runge–Kutta and a time step that automatically shrinks when two bodies come close. Choose one of the classic initial conditions. Switch on "Shadow system" to start a copy of the system with one coordinate changed by a millionth – the graph shows how quickly the two copies part.

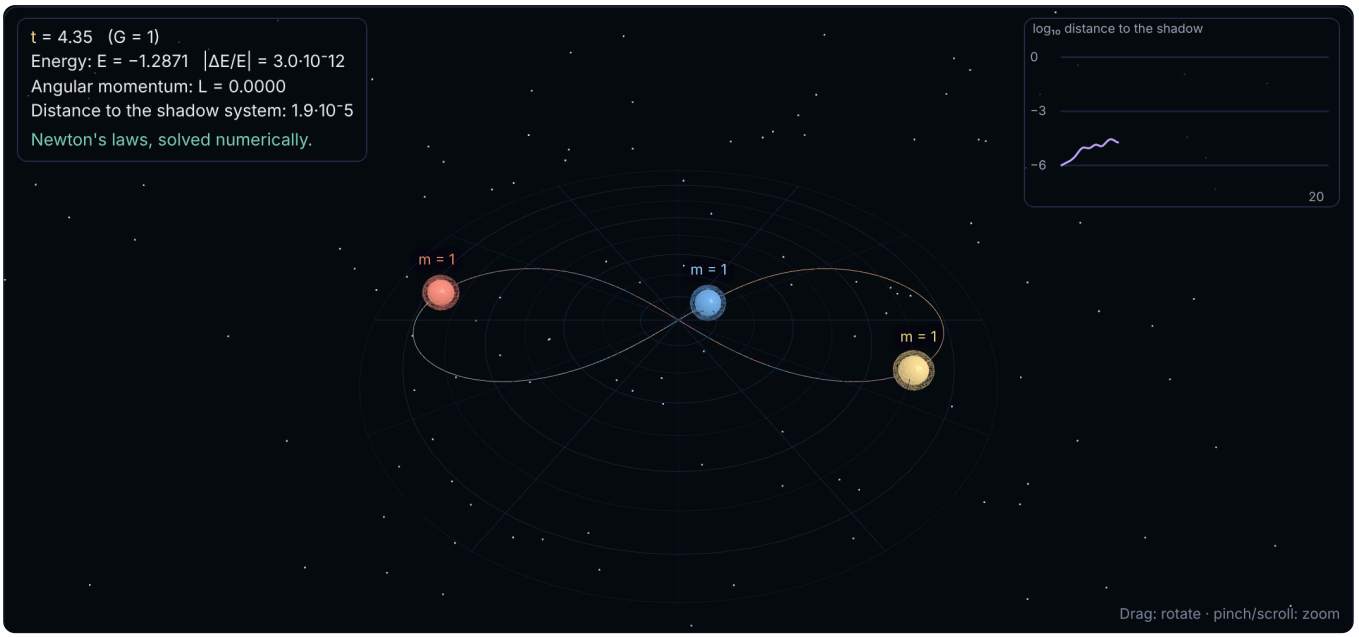


Figure 1. The simulator: release three bodies – snapshot of the interactive simulation.

Interactive version: https://mhgfysik.dk/trelegemeproblemet-en.html#simulator

Try it yourself.

1. Figure eight: does the shadow system keep up? For how long? (The orbit is stable, so the difference grows only slowly.)
2. Lagrange's triangle with three equal masses: how long does the triangle last? What happens when it breaks up?
3. Pythagoras: three masses of 3, 4 and 5 start at rest at the corners of a 3-4-5 triangle. How does it end? Watch the energy and angular momentum in the readout – they should be conserved.
4. Star, planet and moon: why is this system so much more predictable than Pythagoras?

Euler and Lagrange: the five special solutions

Although there is no general solution, there are special solutions in which the figure formed by the three bodies keeps its shape and merely rotates and scales. Leonhard Euler found the **collinear** solutions in 1767: the three bodies always lie on a straight line that rotates. Joseph-Louis Lagrange found the **equilateral triangle** in 1772 in his prize essay "Essai sur le problème des trois corps": for

any distribution of the masses the three bodies can move so that they always form an equilateral triangle. Lagrange himself thought these solutions were mere curiosities.

The Euler and Lagrange solutions with three equal masses are unstable – try them in the simulator. But if one mass is much larger than the other two, and the middle one much larger than the smallest, Lagrange's triangle is stable. Edward Routh showed in 1875 that this requires

$$27\,(m_1m_2 + m_2m_3 + m_3m_1) < (m_1 + m_2 + m_3)^2,$$

for a sun and planet with a small third mass: $\mu = m_2/(m_1+m_2) < 0.0385$. Jupiter has $\mu \approx 0.001$ and satisfies the condition with a wide margin. In 1906 Max Wolf in Heidelberg found the asteroid (588) Achilles 60° ahead of Jupiter in its orbit – the first example in nature of Lagrange's triangle. Today more than 10,000 of these **Trojans** are known, sharing Jupiter's orbit in two groups, and NASA's *Lucy* probe is on its way to visit some of them.

The restricted three-body problem

If the third body is so light that it does not affect the other two – a satellite, an asteroid, a dust grain – the two heavy bodies move in circles about their centre of mass. In a coordinate system rotating with them they stand still, and the light particle feels the gravity of both together with the centrifugal and Coriolis forces. With the distance between the heavy bodies as unit of length and the orbital period 2π as unit of time, they lie at $(-\mu, 0)$ and $(1-\mu, 0)$, and the particle's motion is

$$\ddot{x} - 2\dot{y} = \frac{\partial \Omega}{\partial x}, \qquad \ddot{y} + 2\dot{x} = \frac{\partial \Omega}{\partial y}, \qquad \Omega = \tfrac{1}{2}(x^2+y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}.$$

Carl Jacobi found one conserved quantity in 1836, the **Jacobi constant** $C = 2\Omega - \dot{x}^2 - \dot{y}^2$. Since $\dot{x}^2 + \dot{y}^2 \geq 0$, the particle can only be where $2\Omega \geq C$. The boundary is called the **zero-velocity curve**, and George William Hill used it in 1878 to show that the Moon can never escape from the Earth. The equilibrium points, where $\nabla\Omega = 0$, are the five **Lagrange points**: L1, L2 and L3 on the line (Euler's solutions) and L4 and L5 at the apexes of two equilateral triangles (Lagrange's).

The Lagrange points in 3D

The surface shows the effective potential $-\Omega$ in the rotating frame, with the height magnified near the top so that the small differences between the Lagrange points can be seen. The wells are the two heavy bodies; L1, L2 and L3 are saddle points, and L4 and L5 are hilltops. That L4 and L5 are nevertheless stable is due to the Coriolis force: a particle rolling down the hill is deflected and goes into orbit around the top. Choose a particle and watch its orbit, and the light-green zero-velocity curve for its Jacobi constant.

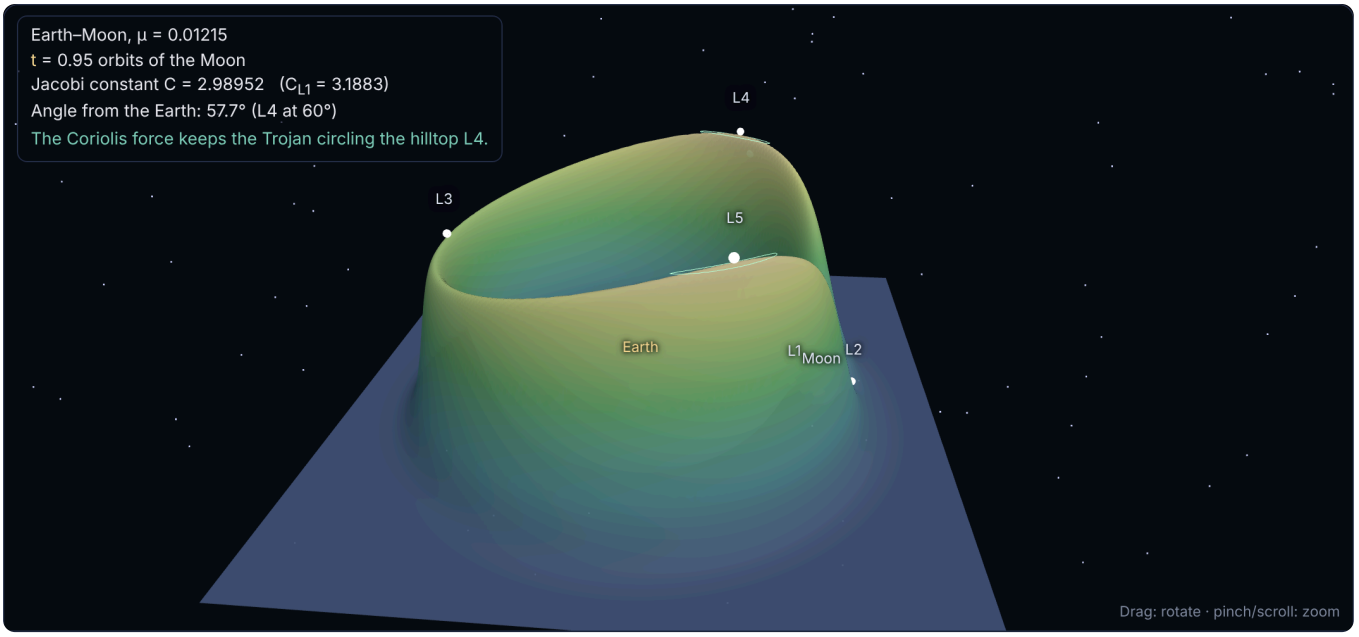


Figure 2. The Lagrange points in 3D – snapshot of the interactive simulation.

Interactive version: https://mhgfysik.dk/trelegemeproblemet-en.html#lagrangepunkter

Try it yourself.

1. The Trojan (Earth–Moon): it starts a little to the side of L4 and circles the point in an elongated "tadpole" orbit. Why does it not roll down the hill?
2. The horseshoe (Sun–Jupiter; here the differences in the landscape are too small to see): the particle shuttles between the L4 and L5 sides and turns at L3. Saturn's moons Janus and Epimetheus swap orbits in this way every four years.
3. L1: how quickly does the particle leave the point? Why must probes at L1 and L2 (SOHO, James Webb) correct their orbits with rocket engines?
4. Earth–Moon: the particle cannot cross the zero-velocity curve. What does that mean for whether it can reach the Moon?

The Lagrange points in practice

Sun–Earth L1 (1.5 million km towards the Sun): the solar observatory SOHO since 1996 – an uninterrupted view of the Sun. **Sun–Earth L2** (1.5 million km away from the Sun): Planck, Gaia and the James Webb telescope, which has the Sun, Earth and Moon in the same direction and can keep cold behind its sunshield. **L4 and L5**: Jupiter's Trojans, the Trojans of Neptune and Mars, and at least two small Earth Trojans (2010 TK_7 and 2020 XL_5).

Where is L1?

For a small mass m_2 at distance a from m_1, L1 and L2 lie at a distance

$$r \approx a\sqrt[3]{\frac{m_2}{3m_1}}$$

from m_2 – the **Hill radius**. For the Earth and the Sun it is about 1.5 million km, four times the Moon's distance. Within the Hill radius the Earth's gravity dominates the Sun's tidal force, and moons can orbit stably there.

Poincaré and the discovery of chaos

In 1885 the Swedish mathematician Gösta Mittag-Leffler announced, on behalf of King Oscar II, a prize for solving the N-body problem: find a convergent series describing the motion for all time. The 34-year-old Henri Poincaré won the prize in January 1889 with a memoir on the restricted three-body problem – without solving it, but with so many new ideas that the judges were convinced. While the memoir was being printed in *Acta Mathematica*, the young Swedish mathematician Lars Edvard Phragmén, who was editing it, raised some questions, and Poincaré discovered a serious error. He had assumed that certain curves in phase space closed neatly; in reality they intersect each other infinitely many times in a web he called *homoclinic* intersections. The printed issue was recalled and destroyed – Poincaré paid himself, and it cost him more than the prize. The corrected version of 1890 became the foundation of his great work *Les méthodes nouvelles de la mécanique céleste* (1892–1899), and in 1908 he summed up in *Science et méthode*:

> *It may happen that small differences in the initial conditions produce very great ones in the final phenomena … Prediction becomes impossible.*

This was the discovery of what we now call **deterministic chaos**: the motion is completely determined by Newton's laws, yet so sensitive to the initial conditions that in practice it cannot be predicted. Try the shadow system in the simulator above with the Pythagorean initial conditions. Poincaré's tool – cross-sections of phase space, now called Poincaré sections – is described on the page on Hamiltonian mechanics.

Sundman's series – the answer that cannot be used

King Oscar's question was in fact answered. In 1912 the Finnish mathematician Karl Frithiof Sundman showed that the three-body problem (when the angular momentum is not zero) has a solution in the form of a convergent power series in $t^{1/3}$ – after a clever rescaling of time – valid for all time. But the series converges so slowly that, according to an estimate by Beloriszky in 1930, about $10^{8\,000\,000}$ terms would be needed to calculate with astronomical accuracy. Chaos does not disappear because one has a formula; it hides in the formula's immense length.

The figure eight and other new orbits

In 1993 the physicist Cristopher Moore found numerically an entirely new periodic solution: three equal masses chasing each other round a figure eight. Alain Chenciner and Richard Montgomery proved in 2000 that it exists. With $G = m = 1$ the initial conditions are (Carles Simó):

$$\mathbf{r}_1 = -\mathbf{r}_2 = (-0.97000436,\ 0.24308753), \quad \mathbf{r}_3 = \mathbf{0}, \qquad \mathbf{v}_3 = (0.93240737,\ 0.86473146), \quad \mathbf{v}_1 = \mathbf{v}_2 = -\tfrac{1}{2}\mathbf{v}_3.$$

In 2013 Milovan Šuvakov and Veljko Dmitrašinović found 13 new families of periodic orbits, and with computer searches thousands are now known. Another classic example is the **Pythagorean problem**, posed by Carl Burrau in 1913: masses 3, 4 and 5 at rest at the corners of a right-angled triangle with sides 3, 4 and 5. Only with Victor Szebehely and C. Frederick Peters' computer calculations in 1967 did it become clear that after a long and chaotic dance the two heaviest form a binary star, and the lightest is flung out.

The three-body problem also gave its name to Liu Cixin's science-fiction novel *The Three-Body Problem* (2008), in which a civilisation lives in a chaotic three-star system. Our nearest neighbours, Alpha Centauri A and B and Proxima Centauri, form a real three-star system – but a hierarchical and therefore far more stable one.

Is the Solar System stable?

The Solar System is an N-body problem, and Newton himself thought that God must intervene from time to time to keep it in order. Laplace and Lagrange showed around 1780 that the planets' mean distances do not change to first order in perturbation theory, and Laplace declared the system stable. Poincaré's work showed that the question was far harder. In 1989 Jacques Laskar calculated that the motion of the inner planets is chaotic, with a Lyapunov time of about 5 million years: an uncertainty of 15 metres in the Earth's position today grows to 150 million km – the whole orbit – within 100 million years. Laskar and Mickaël Gastineau found in 2009, by computing 2501 almost identical solar systems 5 billion years ahead, that in about 1% of them Mercury acquired so large an eccentricity that it could collide with Venus or the Sun. The Solar System, in short, is stable enough – but not predictable for ever.

Exercises

1. Show that three equal masses m at the corners of an equilateral triangle of side a can move in circles about the centre of mass with angular velocity $\omega = \sqrt{3Gm/a^3}$.
2. Show for Euler's line with three equal masses at distance a from the middle one that $\omega = \sqrt{5Gm/(4a^3)}$.
3. Does Sun–Earth–(small mass) satisfy Routh's condition? And Pluto–Charon ($m_2/m_1 \approx 0.12$)?
4. Calculate the Hill radius of the Earth in the Sun's field and of the Moon in the Earth's field. Why does the Moon have no moons?
5. Count conserved quantities: how many degrees of freedom (positions and velocities) does the planar three-body problem have, and how many conserved quantities are there? Compare with

the two-body problem.

6. The Lyapunov time of the inner planets is about 5 million years. By what factor does a small error grow in 100 million years?

Sources and further reading

- J.-L. Lagrange: "Essai sur le problème des trois corps", Prix de l'Académie royale des sciences de Paris, 1772.
- H. Poincaré: "Sur le problème des trois corps et les équations de la dynamique", *Acta Mathematica* 13 (1890) 1–270.
- J. Barrow-Green: *Poincaré and the Three Body Problem*. American Mathematical Society, 1997.
- K. F. Sundman: "Mémoire sur le problème des trois corps", *Acta Mathematica* 36 (1912) 105–179.
- C. Moore: "Braids in classical dynamics", *Physical Review Letters* 70 (1993) 3675; A. Chenciner, R. Montgomery: "A remarkable periodic solution of the three-body problem in the case of equal masses", *Annals of Mathematics* 152 (2000) 881–901.
- M. Šuvakov, V. Dmitrašinović: "Three classes of Newtonian three-body planar periodic orbits", *Physical Review Letters* 110 (2013) 114301.
- V. Szebehely, C. F. Peters: "Complete solution of a general problem of three bodies", *Astronomical Journal* 72 (1967) 876.
- J. Laskar, M. Gastineau: "Existence of collisional trajectories of Mercury, Mars and Venus with the Earth", *Nature* 459 (2009) 817–819.
- R. Montgomery: figure-eight animations and initial conditions, people.ucsc.edu/~rmont.

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